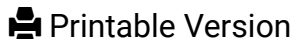


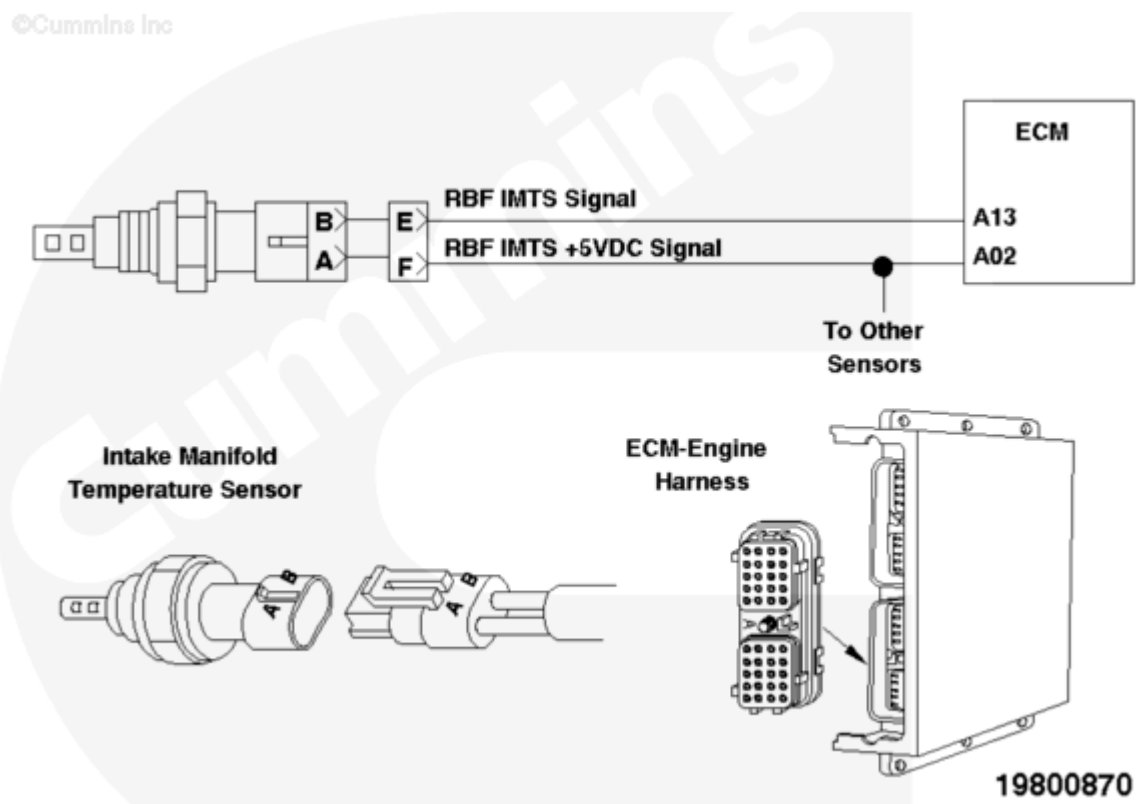
Fault Code: 161

Intake Manifold 3 Temperature Sensor Circuit - Voltage Below Normal or Shorted to Low Source



Overview

Codes	Reason	Effect
Fault Code: 161 PID(P): 105 SPN: 1132 FMI: 4 Lamp: Yellow SRT: 00-699	Intake Manifold 3 Temperature Sensor Circuit - Voltage Below Normal or Shorted to Low Source. Low voltage detected on SIGNAL pin-13 of the main harness A engine control module (ECM) connector.	Fault Code 162 is disabled.



Intake Manifold 3 Temperature Sensor Circuit

Circuit Description

The intake manifold 3 temperature sensor circuit provides the right bank front intake air temperature signal to the ECM. The resistance of the sensor varies with temperature. The ECM detects the change in resistance of the sensor by monitoring the voltage across an internal resistor that is in series with the sensor. The change in voltage across the internal resistor is translated into a temperature change.

Component Location

The intake manifold 3 temperature sensor is located on the right bank front intake manifold. The sensor is located downstream of the aftercooler.

Shoptalk

- The resistance of the sensor varies with the temperature. The reading that you observe should compare to the following table if the sensor is functioning properly.

Temperature	Temperature	Resistance
(°C)	(°F)	(ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Refer to Troubleshooting Fault Code t05-161. (</qs3/pubsys2/xml/en/procedures/81/81-t05-161.html>)